

Vansh

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Summary

I build AI systems that ship. A full-stack ICU sepsis detection platform with SHAP explainability, a multi-document RAG engine with isolated retrieval pipelines, NLP models trained on 400K+ data points. I own products end-to-end: data pipeline to LLM integration to production deployment. If your team needs someone who moves fast, thinks in systems, and gets things done without hand-holding, let's talk.

Technical Skills

AI / ML	Scikit-learn, PyTorch, TensorFlow, XGBoost, Classification, Regression, Neural Networks, Transformers, Fine-tuning
Generative AI & LLMs	LangChain, RAG, Groq, Prompt Engineering, Embeddings, HuggingFace, Agentic Workflows
Backend & APIs	FastAPI, Flask, PostgreSQL, REST APIs, Docker, Git
Data & Explainability	Pandas, NumPy, SHAP, EDA, NLP, Time Series, Matplotlib
Languages	Python, SQL, JavaScript, C++

Projects

PranRakshak AI: Intelligent Hospital Command Center for Early Sepsis Detection [Live Demo](#) | [GitHub](#)

- Architected a real-time sepsis early-warning system on PhysioNet CinC 2019 ICU dataset (40K+ records) using Sepsis-3 clinical criteria; patient priority queue is the core product feature for ICU triage.
- Implemented SHAP-based explainability for per-patient risk drivers, framed as clinical decision support (not diagnosis) to satisfy regulatory constraints.
- Built a RAG clinical copilot using Groq LLMs (Llama 3.3 70B + Llama 4 Scout for OCR) with a strict patient-safety boundary where OCR lab reports never feed the ML model.
- Shipped production full-stack: FastAPI, PostgreSQL, React/Vite frontend, deployed on Render and Vercel.

RAGify: Multi-Document AI Knowledge Assistant [Live Demo](#) | [GitHub](#)

- Designed isolated retrieval pipelines per document to eliminate cross-domain context leakage, a deliberate architectural decision that improves answer fidelity across concurrent multi-document sessions.
- Built semantic retrieval using ChromaDB and HuggingFace embeddings; shipped end-to-end with FastAPI backend and Next.js frontend.

Quora Question Pairs: NLP Semantic Similarity Detection [Live Demo](#) | [GitHub](#)

- Engineered TF-IDF and semantic features across 400K+ question pairs; achieved 79% accuracy and F1 of 0.70 with classical ML. Benchmarked transformer embeddings to quantify the accuracy-latency tradeoff.

Customer Churn Prediction: ANN Classifier [Live Demo](#) | [GitHub](#)

- Trained ANN on structured telecom data achieving 87.73% test accuracy; outperformed Logistic Regression and Random Forest baselines through hyperparameter tuning.

Experience

AI & Machine Learning Intern, EduNet Foundation (AICTE + IBM) Jun 2025 – Aug 2025

- Built and evaluated ML models end-to-end covering data preprocessing, model training, and deployment on real-world datasets within an IBM-structured curriculum.
- Assessed model performance using accuracy, F1-score, and cross-validation across supervised and unsupervised learning tasks.

Education

B.Tech in Computer Science and Engineering, IK Gujral Punjab Technical University CGPA: 8.65 / 10 2023 – 2027

Achievements

- Finalist, Hack On Hills (NIT Hamirpur) | Top 10 Finalist, NIT Jalandhar Hackathon | Top 5 Finalist, LPU Hackathon | 3rd Prize, University Startup Competition

Leadership

- **Founder**, Upstarts, Technology and Startup Club at IKGPTU
- **Class Representative**, CS Department, IKGPTU

Certifications

- [IBM: Fundamentals of Artificial Intelligence](#) | [Anthropic: AI Fluency Certification](#)